INCH-POUND

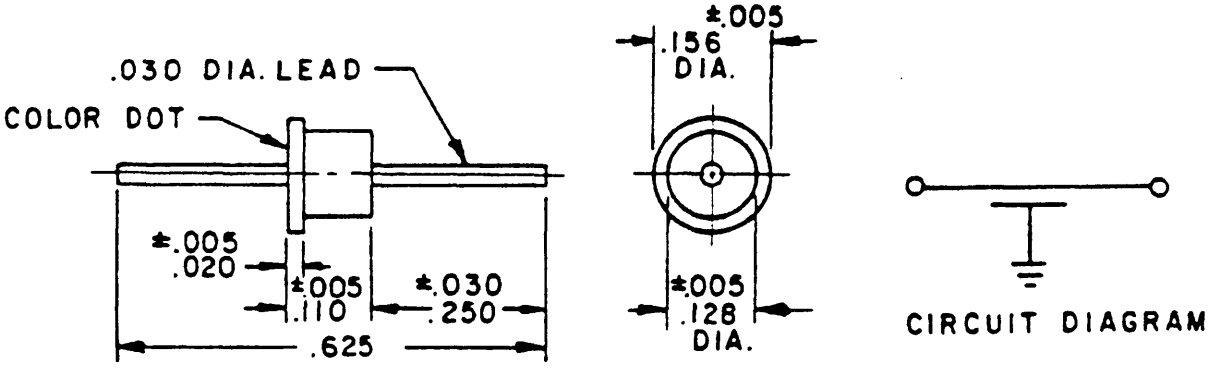
MIL-C-11015/32A(USAF)
15 December 1989
SUPERSEDING
MIL-C-11015/32(USAF)
15 November 1977

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), (FEED-THRU) STYLE CK99

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specification and Standards (DODISS) specified in the solicitation: MIL-C-11015.



APPROXIMATE SIZE

Inches	mm	Inches	mm
.005	0.13	.128	3.25
.020	0.51	.156	3.96
.030	0.76	.250	6.35
.110	2.79	.625	15.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm).

FIGURE 1. Style CK99 capacitors.

(A) denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Ⓐ Design and construction:
Dimensions and configuration - See figure 1.
Case type: Feed-through (solder mount). Solder used internally shall have a minimum melting point of 280°C.
Capacitance value: See table I.
Capacitance tolerance: ±20% (M).
Rated temperature: -55° to +125°C.
Rated voltage: See table I.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage: 200 percent of dc rated voltage applied between the terminal and the mounting flange.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition C (70,000 feet).
Test potential: 200 percent of dc rated voltage applied between the terminal and the mounting flange.

Ⓐ Insulation resistance (IR): Method 302 of MIL-STD-202, condition B, or rated voltage, whichever is less, 500 megohms, minimum, measured between the terminal and the mounting flange.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency: 1.0 ± 0.1 kHz.
Test voltage: 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 5.0%, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition A (50 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G).
Mounting: Capacitors shall be rigidly mounted by the body.

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition A.
IR: 50 megohms, minimum.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition A.

Terminal strength: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.
IR: 50 megohms, minimum.
ΔCap.: Within ±10% of initial measurement.

Solderability: Method 208 of MIL-STD-202 (both ends).

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

Voltage-temperature limits: In accordance with MIL-C-11015 (symbol BW).

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Life (at elevated ambient temperature):
Method 108 of MIL-STD-202.
2,000 hours (for qualification): 250 hours (performance check): 1,750 hours continuation test).
Test potential: 200% of rated voltage.
IR: 100 megohms, minimum (at 85°C);
250 megohms, minimum (at 25°C).
DF: 8.0% maximum (at 25°C).

Marking: Capacitors shall be marked as shown on figure 1. See table 1 for color code. The type designation, capacitance, voltage, and the manufacturer's code or symbol shall be marked on the package container.

TABLE I. Style CK99 characteristics.

Type designation	Capacitance value	Rated voltage	Color dot
	(pF)	(volts, dc)	
CK99BW101M	100	200	Green
CK99BW501M	500	200	Brown
CK99BW122M	1,200	200	White
CK99BW272M	2,700	100	Red
CK99BW502M	5,000	50	Yellow

CONCLUDING MATERIAL

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-F434)

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